



✦ PRIVATE CARE MEMORY FOR INDIAN FAMILIES

कुटुंब Kutumb

*The family care OS that turns daily check-ins, medicines, lab reports,
and AI conversations into a single shared health record.*

VOICE-FIRST • AI-POWERED • PRIVACY BY ARCHITECTURE • BUILT FOR INDIA

18M
Diaspora
Caregivers

\$300B
Adherence Cost /
Yr

6+
Indic
Languages

★ THE BROKEN LOOP

Three lies. Thirty seconds.

"Khaana khaaya?" Haan.

"Tabiyat theek hai?" Bilkul.

"Dawai li?" Le li.

Not because parents want to lie — but because they can't articulate *"my left chest feels heavy when I climb stairs"* on a WhatsApp call. The doctor is 30 km away. The trip itself is exhausting.

THE PATIENT

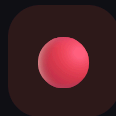
Symptom illiteracy in clinical language — a 68-year-old farmer says *"paer mein jalan"* not "intermittent claudication." That gap is where diagnoses die.

THE FAMILY

18 million Indians abroad. "Surprise emergency flights" cost ₹80k–₹2L each. Almost always preventable with earlier signal.

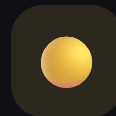
★ THE BROKEN LOOP • CONTINUED

The system is working against everyone.



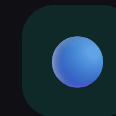
Silent non-adherence

Patients don't say "I stopped my BP meds because they made me dizzy" — they just stop. Six months later: a stroke. Costs **\$300B / year globally.**



10-minute snapshots

A consultation cannot reconstruct 6 months of symptoms. Doctors guess from incomplete narratives. Wrong referrals cost money and the patient time.



Fragmented records

ASHA workers manage 200+ families on paper notebooks. Patient data lives across 8 disconnected silos, owned by none of them.

125,000

Preventable deaths per year in the US alone — directly attributable to medicine non-adherence. Not a developing-world problem. A human problem.

★ THE SOLUTION

One home. One health pattern.

Kutumb is the **layer underneath telemedicine** — the daily, ambient, voice-first companion that catches deterioration months before a crisis.

We're not building a doctor replacement. We're building the **memory and the messenger** between three people who currently don't talk to each other in any structured way.



Voice Check-ins

Hindi, Tamil, English or quick taps — all become structured care events.



Medicine Cabinet

Dose schedules, prescription OCR, missed-dose flags, family escalation.



Lab Report Memory

Upload once. Biomarkers extracted, abnormalities flagged, trends tracked.



Care-Aware AI

Claude answers with the full family context already loaded.



SOS Doctor Handoff

7-day clinical context as a doctor-ready PDF, one tap.

✦ HOW IT WORKS • DAILY LOOP

Three steps. Every single day.

STEP 01

Capture

The day's signals

Voice recording in any Indic language. Photo of a prescription.
Wearable HR / SpO₂. Quick tap on pain scale. All land as structured events in the timeline.

- 🔊 VOICE NLP
- 📄 OCR VISION
- 🕒 WEARABLES

STEP 02

Read

The pattern

Organ scores (Heart · Brain · Gut · Lungs, each 0–100)
make slow changes visible before they become urgent.
Hard rules fire instantly. Claude surfaces trend alerts.

- ❤️ ORGAN SCORES
- 🤖 CLAUDE AI
- ⚡ HARD RULES

STEP 03

Act

As a family

The right person gets the right nudge with enough context to respond. Missed cardiac dose → family escalation. Critical lab → SOS packet auto-generated.

- 🚨 SOS ESCALATION
- 🔔 SMART NUDGES

Four organs. One family wheel.

Each organ score (0–100) is computed daily from check-ins, wearables, lab values, and medicine adherence. The petal extends when you improve.

HEART • हृदय

HR, BP, stress markers, cardiac symptoms

BRAIN • मस्तिष्क

Sleep hours, quality, stress level

GUT • जठर

Bowel, hydration, meal regularity

LUNGS • फुफ्फुस

SpO₂, respiratory symptoms, steps



HEART • हृदय



84



BRAIN • मस्तिष्क



78



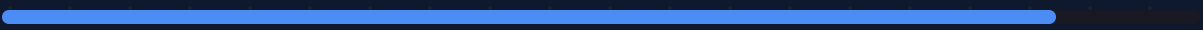
GUT • जठर



72



LUNGS • फुफ्फुस



88

OVERALL • आज

80/100

✦ VOICE-FIRST NLP ENGINE

Speak your language. Any of them.

Rural Indian speech is almost never monolingual. *"Doctor saab ne medicine de di, but pet mein abhi bhi pain hai"* must parse as one utterance, not three failed transcriptions.

HINDI

TAMIL

TELUGU

BENGALI

MARATHI

GUJARATI

KANNADA

+ 4 MORE

PRIMARY: BHASHINI ASR

11 Indic languages · Government-backed · Healthcare domain phrase models · Free for non-commercial use

FALLBACK: WHISPER LARGE-V3

Open weights · On-device TFLite · Works fully offline on 2G

VOICE BIOMARKER LAYER

Pitch variance · Speech rate · Pause distribution → Distress score 0–1. Catches the lying "I'm fine."



Seene mein thodi bhaari feeling hai

KUTUMB AI

I see you've logged chest tightness twice this week. Your BP was 142/88 yesterday. Given your Amlodipine, I'd recommend a doctor visit if this continues.



★ VOICE-FIRST NLP ENGINE

★ MEDICINE SAFETY • CLINICAL GUARDRAILS

Hard rules that can't be overridden.

Safety events are product behavior, not
decoration. These checks fire deterministically
— before Claude even sees the request.
— as one utterance, not three failed transcriptions.

11 PRESCRIPTION CATEGORIES

10 OTC / EMERGENCY CATEGORIES

HINDI • TAMIL • TELUGU • BENGALI • MARATHI • GUJARATI • KANNADA • PUNJABI

PRIMARY: BHASHINI ASP

11 Indic languages • Government-backed • Healthcare domain phrase models • Free for
non-commercial use

FALLBACK: WHISPER LARGE-V3

// Fires instantly, before AI answers

SpO₂ < 88% → SOS trigger

Blood glucose > 300 → Emergency push

Systolic BP > 180 → Crisis flow

NSAID + GI allergy → Block + doctor confirm

3 missed cardiac → Family escalation

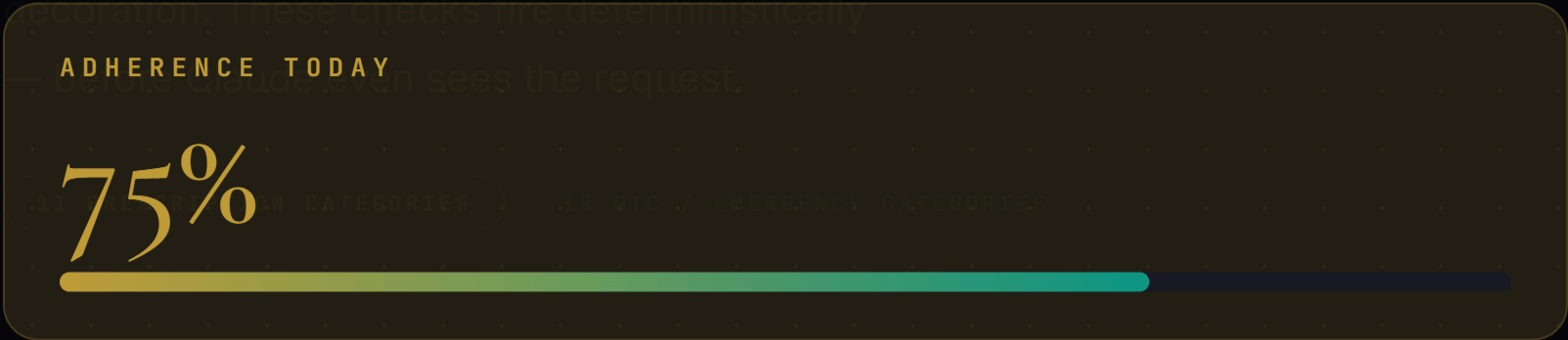
Seene mein thodi bhaari feeling hai

KUTIPPA AI

I see you've logged chest tightness
twice this week. Your BP was 142/88
yesterday. Given your Amlodipine, I'd
recommend a doctor visit if this
continues.

Every dose. Every family member.

Dose schedules, refill windows, prescription
OCR evidence. The whole family's medicine
timeline in one view.



- Metformin 500mg

WITH BREAKFAST • PAPA

08:00

TAKEN
- Amlodipine 5mg

EVENING • PAPA

18:30

OVERDUE
- Salbutamol Inhaler

AS NEEDED • AANYA

PACKED ✓
- Pantoprazole 40mg

NIGHT • DADI

21:00

PENDING
- Atorvastatin 10mg

NIGHT • DADI

21:30

PENDING

✦ PRIVACY · BY ARCHITECTURE

Soft interface. Hard clinical guardrails.

The server never persists health content. The decrypted blob exists in RAM for the duration of one HTTP request, then is purged. DPDP · HIPAA · GDPR compliant by architecture, not by policy promise.

🔒 ON-DEVICE PRIVATE MEMORY

Each user has a single encrypted **.kutumb** file. AES-256-GCM. Key wrapped by biometric via Android Keystore / iOS Secure Enclave. Never uploaded.

⚡ RULES BEFORE GENERATION

Emergency wording, unsafe vitals, medicine conflicts — all checked deterministically before Claude even sees the message.

MEMORY FLOW — ONE SESSION



User speaks
Any Indic language via Bhashini or Whisper



Decrypt .kutumb
Biometric unlock · stays in RAM only



Claude session
Health memory sent in system prompt · 3 outputs: reply · patches · triggers



Re-encrypt locally
Patches applied · saved to device only · never cloud

PRIVACY BY ARCHITECTURE

FAMILY SHARING • कुटुंब

One health record. The whole household.

A child in London sees Dadi's BP trend in Jaipur and Aanya's inhaler status in Mumbai — from one dashboard, without mixing private notes.

Dada • elder

Dadi • elder

Papa • parent

Mummy • parent

You • child

RULES BEFORE GENERATION

Emergency wording, unsafe vitals, medicine conflicts — all checked-deterministically before Claude even sees the message.

BENDRY FLOW, — ONE SESSION

User speaks
Any Indic language via Bhashini or Whisper
PERMISSION LEVELS

full_access View + manage all records, medicines, check-ins, reports

manage_medicines Add / edit schedules, log doses

view_checkins Read-only: check-ins and health scores

receive_sos Emergency alerts and SOS packet only

Re-encrypt locally
Patches applied • saved to device only • never cloud

★ SOS DOCTOR HANDOFF 🇮🇳

One tap. Doctor-ready in seconds.

When something is wrong, the last thing a family should do is hunt for paper prescriptions and old reports. Kutumb has it all ready.

Twilio SMS + push notification to every linked family member with `receive_sos` permission.

● You · child

● Papa · parent

● Mummy · parent



SOS PACKET READY

Papa · 7-day clinical context

PDF READY

● full_access

View + manage all records, medicines, check-ins, reports

- ✓ Current medicines (7) with adherence percentages
- ✓ Vitals log: BP, HR, SpO₂ — last 7 days
- ✓ Latest lab reports + flagged abnormal values
- ✓ Known allergies and contraindications
- ✓ Symptom timeline with severity scores
- ✓ Organ scores + trend direction
- ✓ Active medicines + prescriptions (OCR verified)

● receive_sos

Emergency alerts and SOS packet only

Production-grade. Hackathon-born.

FRONTEND



React Native + Expo SDK 55
Mobile app · TypeScript · 19 screens



Next.js 15 + Tailwind CSS
Web dashboard · TypeScript · 9 routes

BACKEND



FastAPI 1.0 · Python
52 files · 9,568 lines · 14 routers · async Firestore



Firebase Firestore + Auth + Storage
Async client · ID token auth · TTLCache 300s



Docker + Vercel
Dockerfile included · Vercel deployment config ready

API ENDPOINTS • /API/V1/

AUTH

USERS

FAMILY

CHECKINS

MEDICINES

HEALTH

LAB_REPORTS

INSIGHTS

AI/SESSION

EMERGENCY

REFERRALS

WEARABLE

AI/SESSION/ONBOARD

AI/MEMORY

AI & VOICE



Claude claude-sonnet-4-6

Primary AI · Anthropic SDK · health memory in system prompt



Bhashini ASR + Whisper large-v3

Primary + fallback STT · 11 languages · on-device offline



Google Cloud Vision API

Prescription OCR + lab report extraction



WeasyPrint + Jinja2

Doctor referral PDF generation



Twilio SMS

SOS alerts · emergency escalation to family

BY THE NUMBERS

52

PYTHON FILES

14

API ROUTERS

9.5K

LINES OF CODE

19

MOBILE SCREENS



Docker + Vercel

✦ THE PRODUCT • LIVE

Family health. In your pocket.

A native Android + iOS app, paired with a full-featured web dashboard. One Firebase Auth token spans both surfaces. Encrypted local memory works offline.



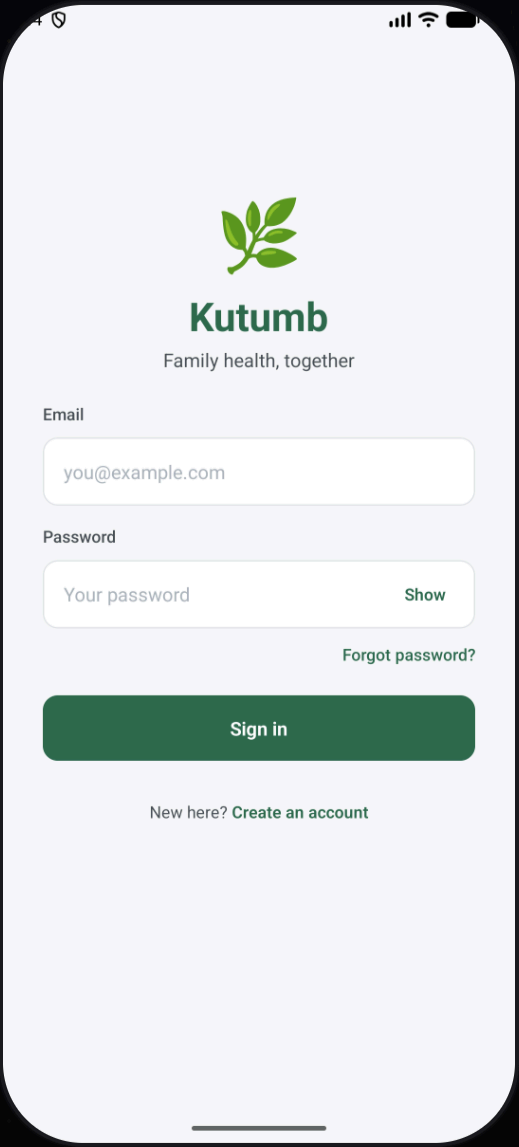
Android APK • available now
kutumb-android-release.apk — built & signed



Web app • Next.js 15
Dashboard • AI chat • Full family board



FastAPI backend • Docker
Dockerfile + Vercel deployment config included



MOBILE • LOGIN SCREEN

React Native • Firebase Auth

Away from home. Close to the heart.

Every Indian abroad — and every Indian whose
parents live in a different city — runs the same broken
loop. Kutumb is the layer they're missing.

18M

Indians living abroad with aging
parents at home — the diaspora
caregiver segment

\$300B

Annual global cost of medicine non-
adherence + 125,000 preventable
deaths/year

800M

Indians with smartphones by 2025,
with zero family health management
apps built for their reality



✦ WHAT'S NEXT • ROADMAP

16 features. One architecture.



Voice Biomarkers

Parkinson tremor, COPD, depression markers from 30s of voice. >85% sensitivity using the user's own baseline.



Pill Identifier

Camera → identify unknown pill → instant contraindication check against current medicines.



Hereditary Risk Tree

Collaborative family medical tree. Auto-adjust alert thresholds based on genetic history.



Maternal & Pediatric

WHO growth charts, National Immunization Schedule sync, danger signs in local language.



Mental Health Mode

Voice-based PHQ-9 / GAD-7 as natural conversation. Crisis detection + escalation to iCall / 988.



Vaccine Tracker

Adult + childhood schedule. Cancer screening reminders by age and gender. Wildly under-utilised.



Caregiver Burnout

Track the diaspora child's stress too. Surface a prompt after 4 weeks of declining sleep + rising care hours.



Scheme Matcher

Auto-match to Ayushman Bharat / PMJAY eligibility. Generate claims documentation automatically.

WHAT'S NEXT? ROADMAP

ROADMAP • CONTINUED



Smart Reminders

Learn when user actually takes medicine. Pair with habit anchor: "When you finish your chai, take your BP pill." +30–45% adherence.



Fall Detection

Accelerometer + gyroscope → 30s grace → if no "I'm okay" voice response → SOS to family + ambulance.



Pharmacy Network

PharmEasy / 1mg routing. Aggregated price comparison. Lab orders to Thyrocare / Dr. Lal PathLabs.



Diet Coach

Region-specific, condition-aware. "Can I eat idli for breakfast?" Photo thali logging with carb + sodium estimate.



Deeper Wearables

Mi Band, boAt, Noise + Omron BP cuffs + Accu-Chek glucometers via Bluetooth. Dominant in India.



Family Health Chat

Group chat inside app. Claude Sunday digest: "Papa missed 2 Metformin this week. Mummy's sleep declining."



Bereavement Mode

Immediately stop traumatising reminders after pregnancy loss or death. One conversation → respectful pause.



Travel Health Mode

Geo-aware hospital routing. Auto-translate Doctor PDF into local language. Vaccination adjustments.



✦ कुटुंब • KUTUMB

Start with one profile. Let the care record grow.

We're not building a doctor replacement. We're building the memory and the messenger between three people who currently don't talk to each other in any structured way.

Request a demo →

View on GitHub

18M

DIASPORA CAREGIVERS

\$300B

ADHERENCE COST / YR

6+

INDIC LANGUAGES

4

ORGAN SCORES